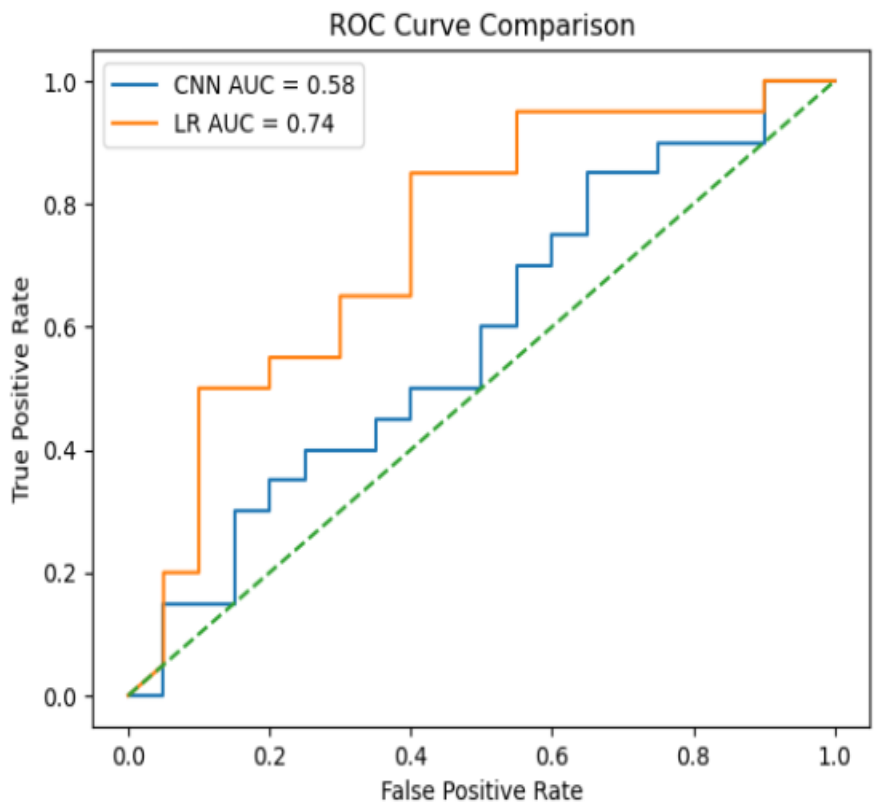


CNN Model Visualization and Performance Analysis Report

This report presents the visual analysis and evaluation results of the Convolutional Neural Network (CNN) developed for food delivery time prediction. The project transformed structured delivery data into synthetic spatial route images and evaluated CNN learning capability using ROC curves, confusion matrices, learning curves, and generated route visualizations.

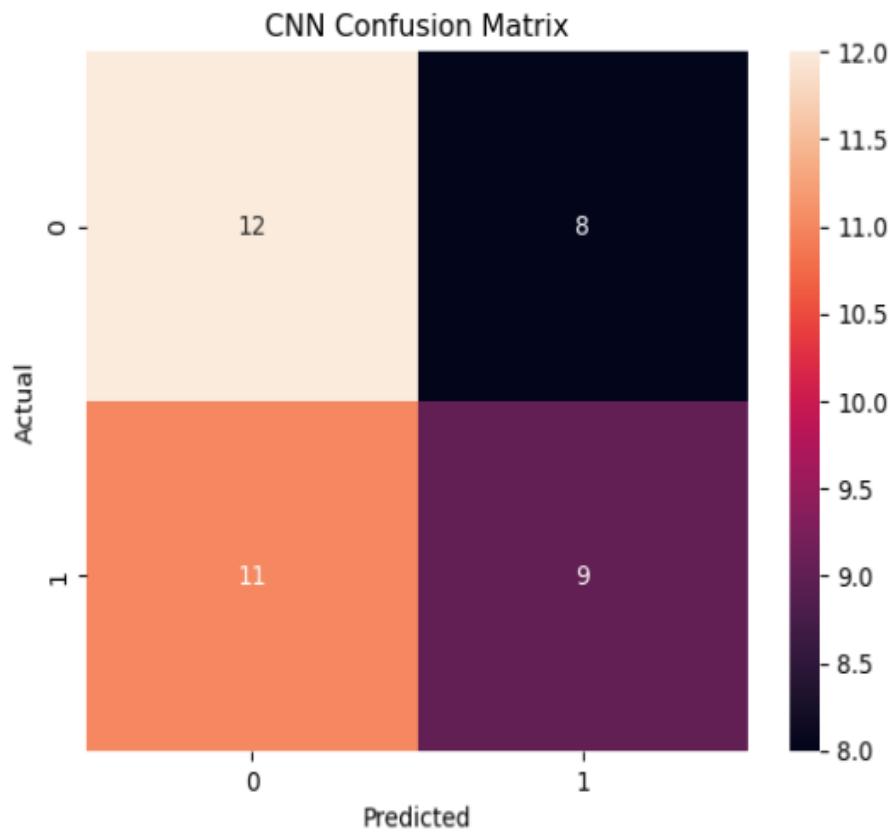
1. ROC Curve Comparison

The ROC (Receiver Operating Characteristic) curve compares the classification performance of the CNN model with Logistic Regression. The CNN achieved an AUC score of 0.58, while Logistic Regression achieved a significantly stronger AUC of 0.74. This indicates that Logistic Regression was more effective at distinguishing between fast and delayed deliveries for the structured delivery dataset. The ROC curve also shows that the CNN learned meaningful patterns from the synthetic images, but its predictive power remained limited compared to traditional machine learning methods.



2. CNN Confusion Matrix Analysis

The confusion matrix provides a detailed breakdown of CNN predictions. The model correctly classified 12 fast deliveries and 9 delayed deliveries. However, the CNN also produced 8 false positives and 11 false negatives. The relatively balanced confusion matrix suggests that the model did not exhibit extreme class bias. However, the number of incorrect classifications highlights the challenges of learning meaningful patterns from simplified synthetic route images.



3. Synthetic Route Image Generation

The delivery dataset was converted into synthetic spatial images to enable CNN-based learning. The generated images contained customer markers, restaurant markers, traffic-based route structures, weather-inspired background textures, and spatial heat regions. Different visual patterns were introduced to help the CNN capture spatial relationships between delivery locations and route conditions. Although these synthetic images improved CNN performance compared to the initial implementation, they still lacked the richness of real-world route maps or satellite imagery.

```
plt.figure(figsize=(10,5))

for i in range(6):

    plt.subplot(2,3,i+1)

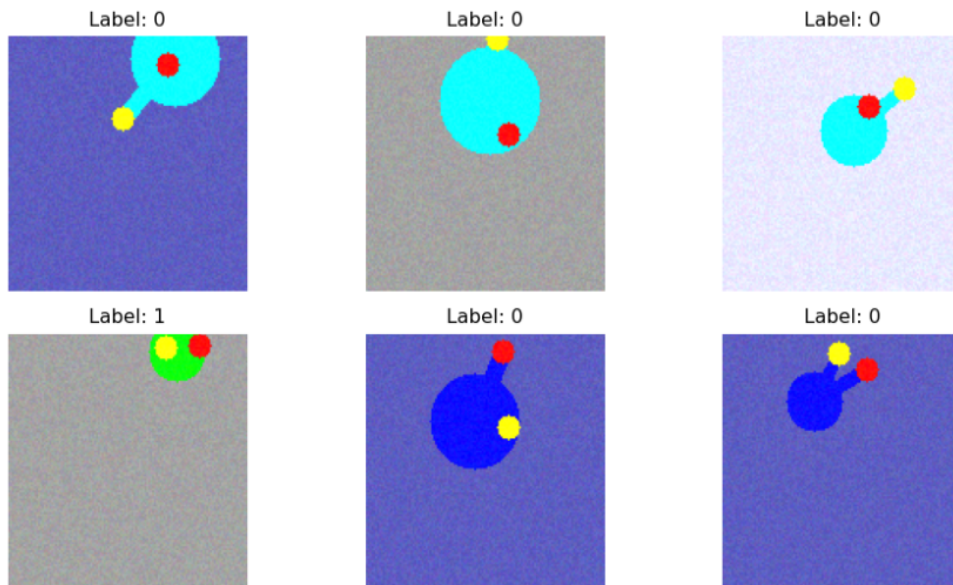
    plt.imshow(X_images[i])

    plt.title(
        f'Label: {y.iloc[i]}'
    )

    plt.axis('off')

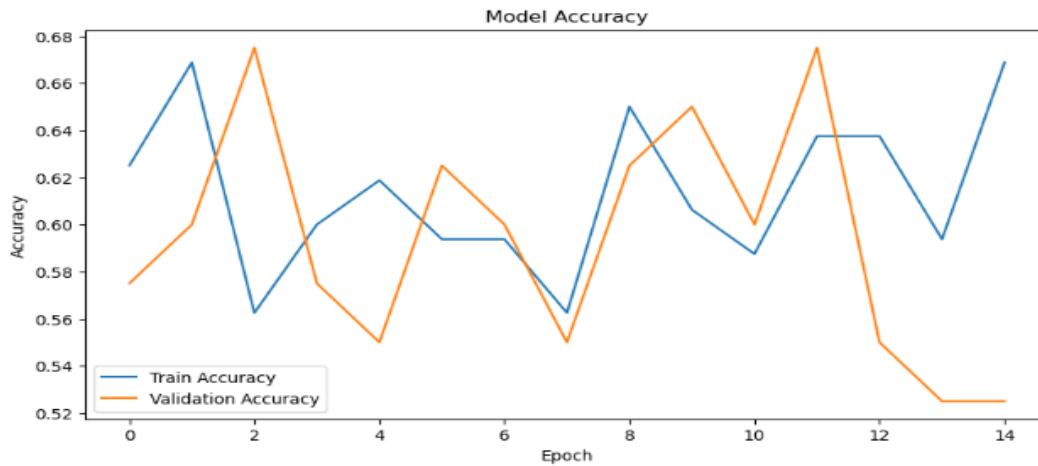
plt.tight_layout()

plt.show()
```



4. Learning Curve Analysis

The learning curves show the training and validation accuracy and loss across epochs. The training and validation accuracy curves remained relatively close throughout training, indicating that the CNN did not suffer from severe overfitting. However, fluctuations in validation accuracy and validation loss suggest instability in generalization performance. This behavior likely resulted from the limited dataset size and simplified synthetic image representations. The curves also suggest mild underfitting because training accuracy itself remained moderate rather than reaching very high values. Regularization methods such as Dropout, BatchNormalization, and data augmentation helped stabilize training behavior.



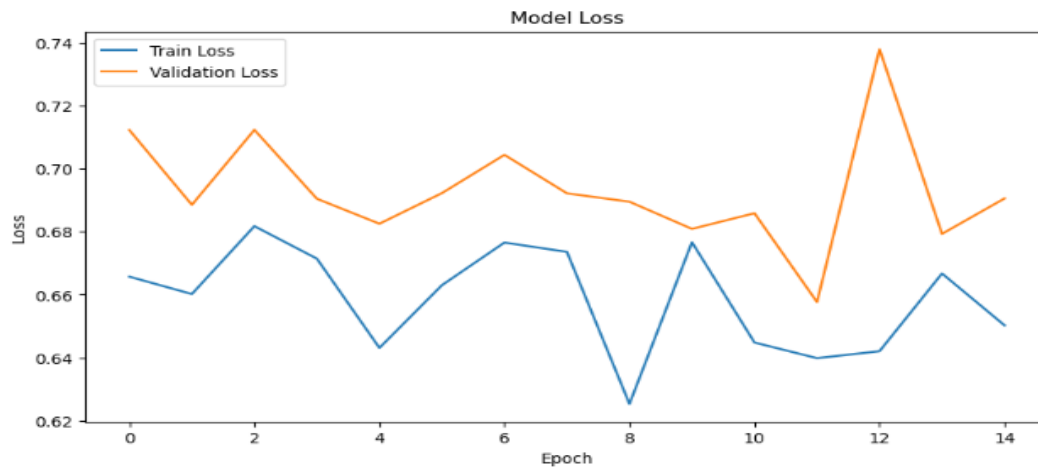
```
3]: plt.figure(figsize=(10,5))

plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])

plt.title('Model Loss')
plt.ylabel('Loss')
plt.xlabel('Epoch')

plt.legend([
    'Train Loss',
    'Validation Loss'
])

plt.show()
```



5. Final Findings and Conclusion

The experimentation demonstrated that CNN performance improved considerably after enhancing the synthetic spatial image representations, applying augmentation strategies, and performing hyperparameter tuning. Key observations include: CNN models learned meaningful spatial patterns from synthetic route images. Cross-validation revealed moderate generalization limitations. Learning curves indicated mild underfitting rather than severe overfitting. Logistic Regression consistently outperformed CNN for the structured delivery dataset. Algorithm suitability is highly dependent on the nature of the data. Overall, the project successfully demonstrated the complete workflow of CNN experimentation, validation, visualization analysis, and model comparison for delivery prediction tasks.